

Morgan Kim

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EDUCATION

Clearwater Institute of Technology | B.S. in Electrical Engineering | Metro City, USA | Sep 2017 - May 2021

EXPERIENCE

Voltline Robotics | Embedded Software Engineer | Metro City, USA | Mar 2023 - Jun 2025

- Improved update speed by 60% by compressing firmware images and resuming interrupted downloads.
- Wrote a hardware-in-the-loop test rig that ran 400 motor-control tests on every firmware build.
- Added a crash logger that saved the last 2 seconds of sensor data to flash, which found the root cause of 5 field failures.
- Sampled the motor's 5 kHz vibration signal at 8 kHz so the fault detector could capture it without aliasing.
- Fixed a race condition on the counter shared by the interrupt handler and the main loop by declaring the counter volatile.
- Rewrote the motor-controller firmware for a warehouse robot fleet of 300 units, cutting control-loop jitter from 120 to 15 microseconds and field motor faults by 60%.

Clearwater Sensors | Hardware Test Engineer | Metro City, USA | Apr 2022 - Feb 2023

- Automated end-of-line tests for a humidity sensor, cutting test time per unit from 90 to 25 seconds across 20,000 units a month.
- Traced a batch of early field failures to a solder-paste change; the findings updated the reflow profile and cut returns from 2.1% to 0.4%.
- Wrote the calibration procedure for the sensor line, cutting calibration drift complaints from 15 to 2 a quarter.
- Designed a test fixture that checked 8 boards at once, raising station throughput from 40 to 150 boards an hour.

PROJECTS

Low-Power Weather Station | Independent Project | C, STM32 | Jan 2025 - Present

- Engineered a next-generation, innovation-driven IoT solution that redefines sustainable environmental sensing.
- Cut average current from 4.2 mA to 0.9 mA by sleeping between readings and batching radio transmissions.
- Published the schematics and firmware openly, and a local school built two stations from them.

Formula Student Electric Car | Electronics Lead, Team of 12 | C | Sep 2019 - May 2021

- Worked on firmware and testing for the robot fleet's motor controllers.
- The CAN-bus firmware for the dashboard was written and tested against the motor controller before each race.
- Rewrote motor-control firmware for a fleet of about 300 robots, cutting control-loop jitter eightfold and motor faults by more than half.

SKILLS

Tools: C, C++, STM32, oscilloscopes, FPGA

Methods: firmware, hardware-in-the-loop testing, CAN, low-power design